

177. Inner Forward Error Correction Sublayer (Inner FEC), type 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R

177.1 Overview

177.1.1 Scope

This clause specifies the Inner Forward Error Correction (Inner FEC) sublayer for some 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R Physical Layer implementations that use 200 Gb/s per lane signaling.

When necessary to differentiate the Inner FEC defined in this clause from the 800GBASE-LR1 Inner FEC defined in Clause 184, the terms 200GBASE-R Inner FEC, 400GBASE-R Inner FEC, 800GBASE-R Inner FEC, and 1.6TBASE-R Inner FEC are used. Within this clause the term Inner FEC refers specifically to 200GBASE-R Inner FEC, 400GBASE-R Inner FEC, 800GBASE-R Inner FEC, and 1.6TBASE-R Inner FEC. Annex 177A provides test vectors constructed using the methods specified in this clause.

177.1.2 Relationship of the Inner FEC to other standards

Figure 177–1 shows the relationship of the Inner FEC sublayer, along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model.

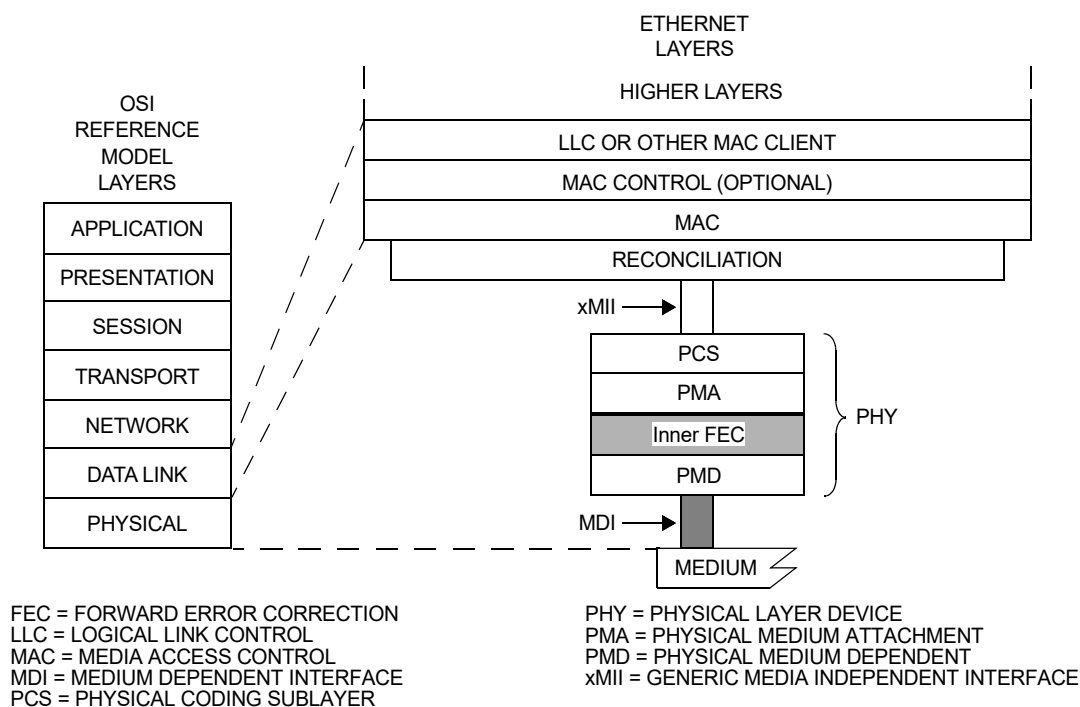


Figure 177–1—Relationship of Inner FEC to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

177.1.3 Summary of functions

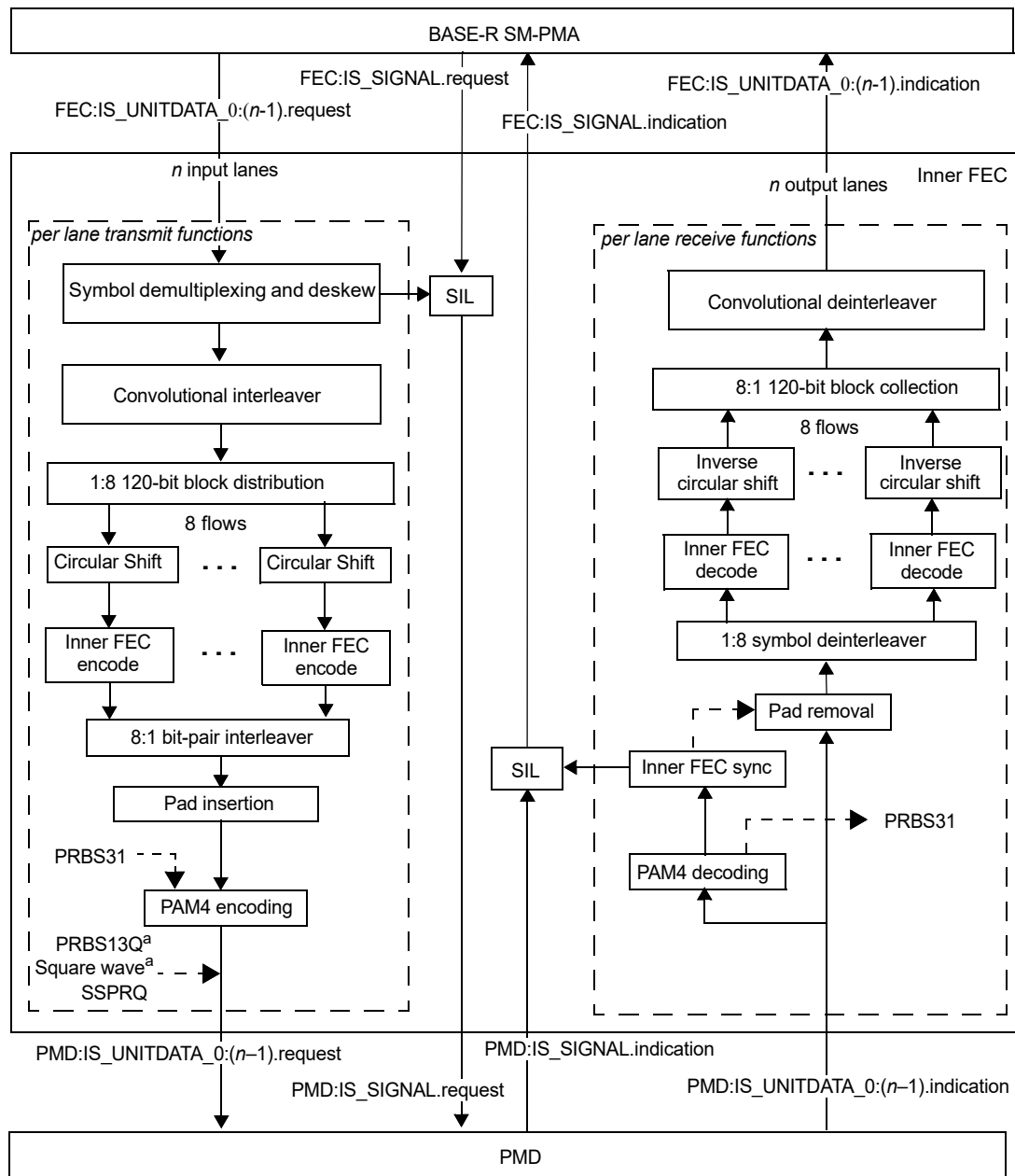
The Inner FEC provides a second layer of FEC protection without decoding the data encoded by the RS-FEC in the PCS.

The Inner FEC provides the following services:

- Interleaving (deinterleaving) the data from (to) the BASE-R SM-PMA using a convolutional interleaver (deinterleaver).
- Distributing (collecting) the convolutional interleaved data to (from) eight Inner FEC flows
- Performing circular shift and inverse circular shift on each Inner FEC flow
- Per Inner FEC flow, a binary(128,120) encoding and decoding
- 8:1 interleaving (1:8 deinterleaving) the eight Inner FEC flows to (from) a single flow
- Pad insertion and removal
- PAM4 encoding and decoding

177.1.4 Functional block diagram

Figure 177–2 provides a functional block diagram of the Inner FEC. The number (n) of input and output lanes matches the number of Inner FEC instances, and this number is 1 for 200GBASE-R, 2 for 400GBASE-R, 4 for 800GBASE-R, and 8 for 1.6TBASE-R.



^a Optional

Figure 177-2—Functional block diagram

177.2 Inner FEC service interface

For the 200GBASE-R Inner FEC, the client sublayer is a 200GBASE-R 8:1 SM-PMA or a 200GBASE-R 1:1 SM-PMA. For the 400GBASE-R Inner FEC, the client sublayer is a 400GBASE-R 16:2 SM-PMA or a 400GBASE-R 2:2 SM-PMA. For the 800GBASE-R Inner FEC, the client sublayer is an 800GBASE-R 32:4 SM-PMA or an 800GBASE-R 4:4 SM-PMA. For the 1.6TBASE-R Inner FEC, the client sublayer is a 1.6TBASE-R 16:8 SM-PMA or a 1.6TBASE-R 8:8 SM-PMA.

The Inner FEC service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-R and 400GBASE-R, in 169.3 for 800GBASE-R, and in 174.3 for 1.6TBASE-R.

The Inner FEC service interfaces are illustrated in Figure 177–2. The service interface primitives are summarized as follows:

```
FEC:IS_UNITDATA_i.request(tx_symbol)
FEC:IS_UNITDATA_i.indication(rx_symbol)
FEC:IS_SIGNAL.request(SIGNAL_OK)
FEC:IS_SIGNAL.indication(SIGNAL_OK)
```

The primitives are defined for $i = 0$ to $n - 1$. In the transmit direction, the Inner FEC sublayer receives n parallel bit streams, each at a nominal rate of 212.5 Gb/s, from the client sublayer. In the receive direction, the Inner FEC sublayer sends n parallel bit streams, each at a nominal rate of 212.5 Gb/s, to the client sublayer. The number of parallel streams, n , is 1 for 200GBASE-R PHYs, 2 for 400GBASE-R PHYs, 4 for 800GBASE-R PHYs, and 8 for 1.6TBASE-R PHYs. Each instance of tx_symbol and rx_symbol takes on one of two values: 0 or 1.

The Inner FEC sublayer receives signal status information from the client sublayer (PMA) using the FEC:IS_SIGNAL.request primitive. The SIGNAL_OK parameter takes one of four values: OK, READY, IN_PROGRESS, or FAIL. When the value of SIGNAL_OK is READY, IN_PROGRESS, or FAIL, the corresponding tx_symbol parameters of the FEC:IS_UNITDATA_i.request primitive on all lanes are unspecified.

The Inner FEC sublayer provides signal status information to the client sublayer (PMA) using the FEC:IS_SIGNAL.indication primitive. The SIGNAL_OK parameter is set to one of four values by the signal indication logic (SIL) according to Table 177–1. When the value of SIGNAL_OK is READY, IN_PROGRESS or FAIL, the corresponding rx_symbol parameters of the FEC:IS_UNITDATA_i.indication primitive on all lanes are unspecified.

Table 177–1—FEC:IS_SIGNAL.indication(SIGNAL_OK) generation

PMD:IS_SIGNAL.indication SIGNAL_OK value	all_synced (see 177.7.2.1)	FEC:IS_SIGNAL.indication SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

In addition to the primitives noted above, an associated clock is transferred from input to output along with the IS_UNITDATA primitives in the transmit and receive direction.

177.3 Service interface below the Inner FEC

The service interface below the Inner FEC is the PMD service interface for the PHY.

The PMD provides signal status information to the Inner FEC using the PMD:IS_SIGNAL.indication(SIGNAL_OK) primitive. The SIGNAL_OK parameter takes one of four values: OK, READY, IN_PROGRESS, or FAIL. When SIGNAL_OK is IN_PROGRESS or FAIL, the rx_symbol parameters are undefined.

The Inner FEC sublayer provides signal status information to the PMD using the PMD:IS_SIGNAL.request(SIGNAL_OK) primitive (see Figure 177–2). The SIGNAL_OK parameter is set to one of four values by the SIL according to Table 177–2. When the value of SIGNAL_OK is READY, IN_PROGRESS or FAIL, the corresponding tx_symbol parameters on all lanes is unspecified.

Table 177–2—PMD:IS_SIGNAL.request(SIGNAL_OK) generation

FEC:IS_SIGNAL.request SIGNAL_OK value	all_locked_demux (see 176.4.4.2.1)	PMD:IS_SIGNAL.request SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

177.4 Transmit function

The following processes are performed independently on each Inner FEC service interface input lane.

177.4.1 Symbol demultiplexing and deskew

The symbol demultiplexing and deskew function is illustrated in Figure 177–3.

The alignment marker lock process within the symbol demultiplexing function determines the RS-FEC symbol-pair, symbol-quartet and codeword boundaries, as described in 176.4.3.2. Alignment marker lock is required for 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R Inner FEC. Note that for 200GBASE-R and 400GBASE-R, the data stream is not altered.

Deskew aligns the codeword boundaries on all PCS lanes within an Inner FEC lane. Deskew is required for 800GBASE-R and 1.6TBASE-R Inner FEC. After deskew, the PCS lanes are recombined by the symbol multiplexing function.

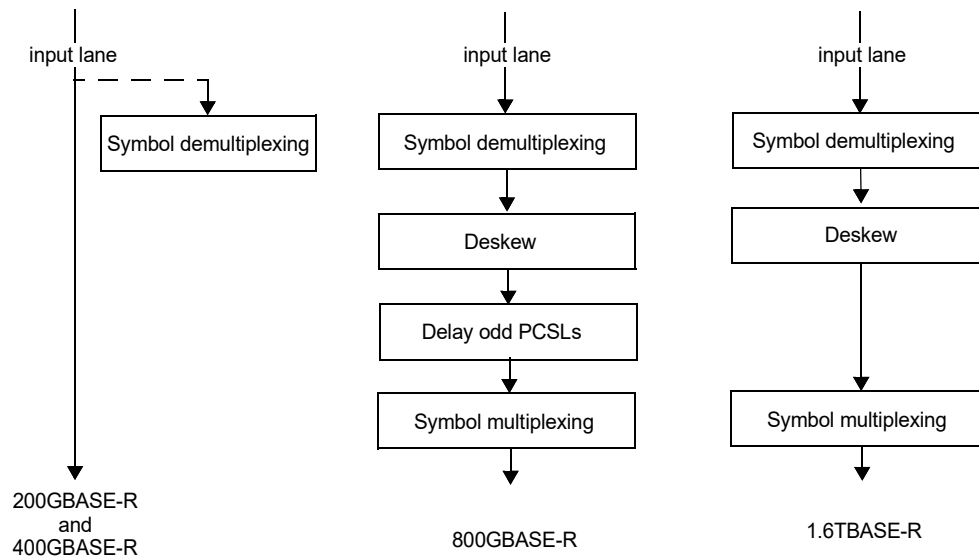


Figure 177-3—Symbol demultiplexing and deskew block diagrams

177.4.1.1 Symbol demultiplexing

The symbol demultiplexing function is performed as defined in 176.4.3.2, with the exception that it operates on the Inner FEC service interface input lanes.

177.4.1.2 Deskew

For 800GBASE-R PHYs, after alignment marker lock is achieved on each of the eight PCSs in an input stream, Skew between PCSs is removed as defined in 172.2.5.1, with the exception that the maximum Skew to be removed is the Skew at SP1 (see Table 169-5) plus the Skew added by the PMA above the Inner FEC.

For 1.6TBASE-R PHYs, after alignment marker lock is achieved on each of the two PCSs in an input stream, Skew between PCSs is removed as defined in 175.2.5.1, with the exception that the maximum Skew to be removed is the Skew at SP1 (see Table 174-5) plus the Skew added by the PMA above the Inner FEC.

NOTE— In 200GBASE-R and 400GBASE-R PHYs, deskew is not required because the SM-PMA above the Inner FEC already removes the Skew by aligning the PCS lanes within each PMA lane to a 4-codeword boundary.

177.4.1.3 Delay odd PCS lanes

Delay of odd PCS lanes is performed only by the 800GBASE-R PHYs. The odd PCSs are delayed by one RS-FEC symbol (10 bits), as illustrated in Figure 176-4, to remove the checkerboard pattern created by the distribution of RS-FEC symbols to PCS lanes.

177.4.1.4 Symbol multiplexing

For 800GBASE-R PHYs, after aligning the PCS lanes to the alignment markers and delaying the odd PCSs, eight PCSs are re-combined into a single data stream using the symbol-pair multiplexing function described in 176.4.2.5.1.

For 1.6TBASE-R PHYs, after aligning the PCSs to the alignment markers, two PCSs are re-combined into a single data stream using the symbol-quartet multiplexing function described in 176.4.2.5.2.

177.4.2 Convolutional interleaver

A convolutional interleaver shall be used on each deskewed Inner FEC lane to systematically distribute RS-FEC symbols from 12 different RS-FEC codewords into an Inner FEC codeword, in order to improve robustness of the system to bursts of errors. Using the RS-FEC symbol-quartet boundaries found by the symbol demultiplexing and deskew process (see 177.4.1) the convolutional interleaver creates groups of four RS-FEC symbols from four RS-FEC codewords. When using PRBS31 encoded by the Inner FEC test mode (see 177.6.1.1), the selection of the RS-FEC symbol-quartet boundary position is unspecified.

The convolutional interleaver contains three delay lines. The first line (Delay Line 0) delays the data by $2 \times Q \times 4$ RS-FEC symbols, the second line (Delay Line 1) by $1 \times Q \times 4$ RS-FEC symbols, and the last line (Delay Line 2) adds no delay. The values of Q are shown in Table 177–3.

Table 177–3—Configuration of convolutional interleaver delay lines

Ethernet rate	Value of Q
200GBASE-R	192
400GBASE-R	96
800GBASE-R	48
1.6TBASE-R	24

Data from the deskewed PMA lane is fed into each delay line one RS-FEC symbol-quartet at a time. The input data round-robins between the three delay lines in the order Delay Line 0, then Delay Line 1 and lastly Delay Line 2.

The output of the convolutional interleaver round-robins between the three delay lines, receiving one RS-FEC symbol-quartet from each at a time, in the order Delay Line 0, then Delay Line 1 and lastly Delay Line 2.

Figure 177–4 illustrates the delay lines of the convolutional interleaver using Q instances of a RS-FEC symbol-quartet sized base delay element labeled “D”. The input and output switches are always aligned to the same row.

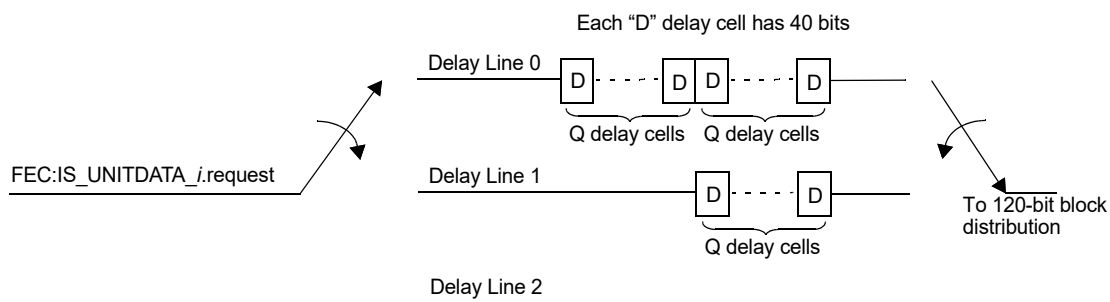


Figure 177–4—Convolutional interleaver

177.4.3 120-bit block distribution

The output of the convolutional interleaver is distributed to 8 Inner FEC flows, in blocks of 120 bits (12 RS-FEC symbols, or 3 RS-FEC symbol-quartets) in a round-robin fashion from flow 0 to 7. The first RS-FEC symbol-quartet within a 120-bit block is always from delay line 0. Within each RS-FEC symbol, bit 0 is transmitted first and bit 9 is transmitted last.

177.4.4 Circular shift

The circular shift function enhances the burst error tolerance of the Inner FEC. The 12 RS-FEC symbols within each 120-bit block are shuffled to maximize the distance between transmitted PAM4 symbols from the same RS-FEC codeword. The circular shift function, $CS(n)$, circularly shifts the 12 RS-FEC symbols by n RS-FEC symbols towards the last arrived RS-FEC symbol. The circular shift function applied to each Inner FEC flow is shown in Figure 177–5, where symbol A is the first arrived RS-FEC symbol in each 120-bit block from the convolutional interleaver.

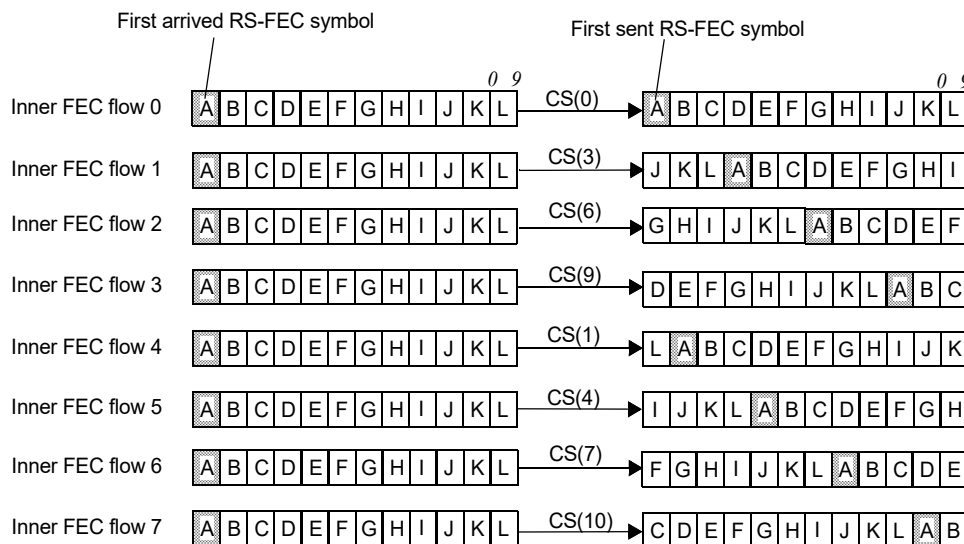


Figure 177–5—Circular shift

177.4.5 Inner FEC encode

In each of the 8 Inner FEC flows, the 120-bit output block of the circular shift, $m\langle 119:0 \rangle$, shall be encoded by a systematic binary(128,120) code, appending 8 parity bits at the end of each 120-bit block to produce a 128-bit Inner FEC codeword $c\langle 127:0 \rangle$. $m(119)$ is the first bit received from the circular shift and $c(127)$ is the first bit transmitted after Inner FEC encode.

The encode process is illustrated in Figure 177–6. First, each pair of bits to be modulated into a PAM4 symbol are XORed resulting in 60 bits, $m_xor\langle 59:0 \rangle$, from the 120-bit message.

For $k = 0:59$

$$m_xor(k) = m(2k) \text{ XOR } m(2k+1)$$

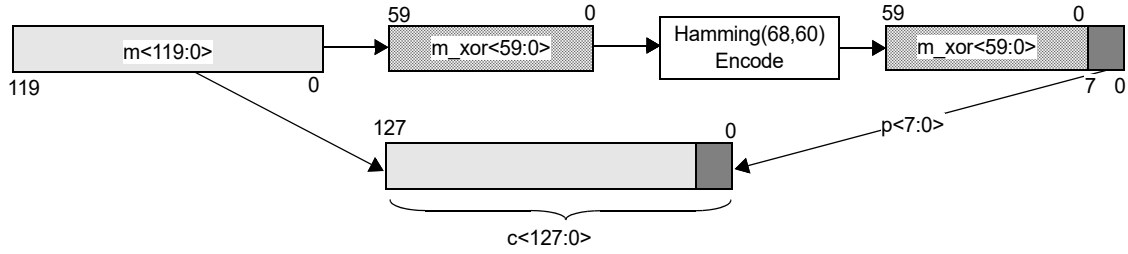


Figure 177-6—Inner FEC encode process

Next, a Hamming (68,60) encoder is used to generate 8 parity bits based on the 60 bits from the previous step, $m_xor<59:0>$.

The systematic Hamming code is defined in terms of its parity-check matrix. Consider the function g which maps an integer i , $0 \leq i \leq 67$, to a binary column vector as shown in Equation (177-2). α is a primitive element in Galois Field $GF(2^7)$, and the element α^i can be expressed as a linear combination of $\alpha^0, \alpha^1, \alpha^2, \alpha^3, \alpha^4, \alpha^5, \alpha^6$, as shown in Equation (177-3). $(s_{0,i}, s_{1,i}, s_{2,i}, s_{3,i}, s_{4,i}, s_{5,i}, s_{6,i})$ are the binary coefficients of the linear combination that creates the element α^i in the Galois Field $GF(2^7)$ with primitive polynomial $x^7 + x^3 + 1$.

The parity-check matrix H is then a 8×68 binary matrix as shown in Equation (177-4). To obtain the encoder matrix G , we calculate P and B using Equation (177-5) and Equation (177-6). The generator matrix of the Hamming(68,60) code is $G = [I_{60}; G_{60,8}]$, where I_{60} is a 60×60 identity matrix, and $G_{60,8} = P^T$, where the superscript T denotes a matrix transpose operator, is a 60×8 matrix used to generate the 8 parity bits as given in Table 177-4. Within each row the MSB is on the left.

The eight parity bits are calculated as

$$p<7:0> = m_xor<59:0> \cdot G_{60,8} \quad (177-1)$$

where the “ $\cdot$ ” denotes a matrix dot multiplication in Equation (177-1) and Equation (177-5). The matrix dot multiplication in these equations includes an XOR operation as the addition step.

The output codeword $c<127:0>$ is:

$$\begin{aligned} c<127:8> &= m<119:0> \\ c<7:0> &= p<7:0> \end{aligned}$$

$$g(i) = \begin{bmatrix} \alpha^i \\ 1 \end{bmatrix} = \begin{bmatrix} s_{0,i} \\ s_{1,i} \\ s_{2,i} \\ s_{3,i} \\ s_{4,i} \\ s_{5,i} \\ s_{6,i} \\ 1 \end{bmatrix} \quad (177-2)$$

$$\alpha^i = s_{0,i} \times \alpha^0 + s_{1,i} \times \alpha^1 + s_{2,i} \times \alpha^2 + s_{3,i} \times \alpha^3 + s_{4,i} \times \alpha^4 + s_{5,i} \times \alpha^5 + s_{6,i} \times \alpha^6 \quad (177-3)$$

$$\mathbf{H} = [g(0), g(1), g(2), \dots, g(66), g(67)] \quad (177-4)$$

$$\mathbf{P} = \mathbf{B} \cdot [g(0), g(1), g(2), \dots, g(58), g(59)] \quad (177-5)$$

$$\mathbf{B} = [g(60), g(61), \dots, g(66), g(67)]^{-1} \quad (177-6)$$

The superscript “−1” denotes the inverse of the matrix in Equation (177–6).

Table 177–4—Generator matrix for Hamming(68,60)^a

G(0:11)	G(12:23)	G(24:35)	G(36:47)	G(48:59)
1,0,0,1,0,1,0,0	1,0,1,0,1,0,0,0	0,0,1,1,0,1,1,1	1,1,1,0,0,0,0,0	0,1,1,1,1,0,0,1
0,1,0,0,1,0,1,0	0,1,0,1,0,1,0,0	1,1,0,0,0,0,1,0	0,1,1,1,0,0,0,0	1,1,1,0,0,1,0,1
0,0,1,0,0,1,0,1	0,0,1,0,1,0,1,0	0,1,1,0,0,0,0,1	0,0,1,1,1,0,0,0	1,0,1,0,1,0,1,1
1,1,0,0,1,0,1,1	0,0,0,1,0,1,0,1	1,1,1,0,1,0,0,1	0,0,0,1,1,1,0,0	1,0,0,0,1,1,0,0
1,0,1,1,1,1,0,0	1,1,0,1,0,0,1,1	1,0,1,0,1,1,0,1	0,0,0,0,1,1,1,0	0,1,0,0,0,1,1,0
0,1,0,1,1,1,1,0	1,0,1,1,0,0,0,0	1,0,0,0,1,1,1,1	0,0,0,0,0,1,1,1	0,0,1,0,0,0,1,1
0,0,1,0,1,1,1,1	0,1,0,1,1,0,0,0	1,0,0,1,1,1,1,0	1,1,0,1,1,0,1,0	1,1,0,0,1,0,0,0
1,1,0,0,1,1,1,0	0,0,1,0,1,1,0,0	0,1,0,0,1,1,1,1	0,1,1,0,1,1,0,1	0,1,1,0,0,1,0,0
0,1,1,0,0,1,1,1	0,0,0,1,0,1,1,0	1,1,1,1,1,1,1,0	1,1,1,0,1,1,1,1	0,0,1,1,0,0,1,0
1,1,1,0,1,0,1,0	0,0,0,0,1,0,1,1	0,1,1,1,1,1,1,1	1,0,1,0,1,1,1,0	0,0,0,1,1,0,0,1
0,1,1,1,0,1,0,1	1,1,0,1,1,1,0,0	1,1,1,0,0,1,1,0	0,1,0,1,0,1,1,1	1,1,0,1,0,1,0,1
1,1,1,0,0,0,1,1	0,1,1,0,1,1,1,0	0,1,1,1,0,0,1,1	1,1,1,1,0,0,1,0	1,0,1,1,0,0,1,1

^a Within each row, the MSB is on the left.

177.4.6 8:1 bit-pair interleaving

After Inner FEC encoding, the eight flows of Inner FEC codewords shall be multiplexed together to form a single output lane. The 8:1 multiplexing is performed from flow 0 to flow 7, in a round-robin fashion, on pairs of bits beginning with c<127:126>. Each pair of bits will later be modulated into a PAM4 symbol.

177.4.7 Pad insertion and format

A 1024-bit pad shall be inserted after every 8704 Inner FEC codewords (see Figure 177–7). The nominal rate for each Inner FEC lane after pad insertion is 226.875 Gb/s. The ratio between the rate before and after pad insertion is 1088/1089. The bit rate after pad insertion is 363/340 of the bit rate of the tx_symbol stream at the Inner FEC service interface.

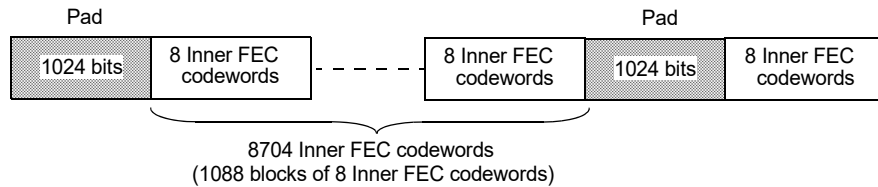


Figure 177–7—Pad insertion interval

The 1024-bit pad is equivalent to Inner FEC encoding (see 177.4.5) and 8:1 bit-pair interleaving (see 177.4.6) the $8 \times 120b$ data provided in Table 177–5, when Inner FEC flow i is fed pad_cw_ i ($i=0:7$) (see Figure 177–8).

Table 177–5—Pad format

	Pad frame alignment sequence $\text{pad_cw}_i<127:122>$, $\text{m}_i<119:114>$	Pad bits, $\text{pad_cw}_i<121:8>$, $\text{m}_i<113:0>$	Parity $\text{pad_cw}_i<7:0>$ $\text{p}_i<7:0>$
pad_cw_0	01 01 10	See 177.4.7.2	Calculated based on $\text{pad_cw}_0<127:8>$
pad_cw_1	01 10 10	See 177.4.7.2	Calculated based on $\text{pad_cw}_1<127:8>$
pad_cw_2	10 01 11	See 177.4.7.2	Calculated based on $\text{pad_cw}_2<127:8>$
pad_cw_3	01 00 01	See 177.4.7.2	Calculated based on $\text{pad_cw}_3<127:8>$
pad_cw_4	01 10 10	See 177.4.7.2	Calculated based on $\text{pad_cw}_4<127:8>$
pad_cw_5	01 10 01	See 177.4.7.2	Calculated based on $\text{pad_cw}_5<127:8>$
pad_cw_6	00 01 10	See 177.4.7.2	Calculated based on $\text{pad_cw}_6<127:8>$
pad_cw_7	10 10 11	See 177.4.7.2	Calculated based on $\text{pad_cw}_7<127:8>$

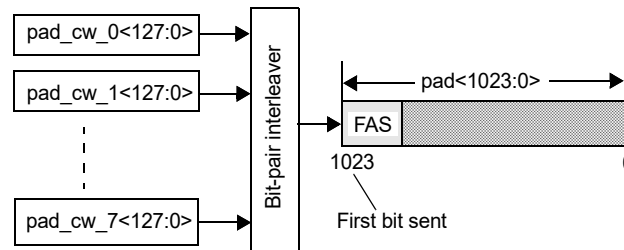


Figure 177–8—Pad codewords mapping

177.4.7.1 Pad frame alignment sequence

The Frame Alignment Sequence (FAS) is a fixed pattern that is the first 48 bits transmitted in each pad and enables the receiver to locate the pad. The fixed FAS pattern is as follows with the leftmost bit transmitted first: 01011001 01010010 01100100 10100110 10101101 10011011.

177.4.7.2 Pad bits

Pad bits spanning across all 8 pad codewords, as shown in Table 177–5, are filled with 0 and then scrambled using a self-synchronizing scrambler using the polynomial defined in Equation (94–3) and shown in Figure 177–9. The scrambler runs sequentially on the pad bits from pad_cw_0 through pad_cw_7 , as shown in Figure 177–10. The initial state of the scrambler is any non-zero state, and the scrambler state is retained from the previous pad.

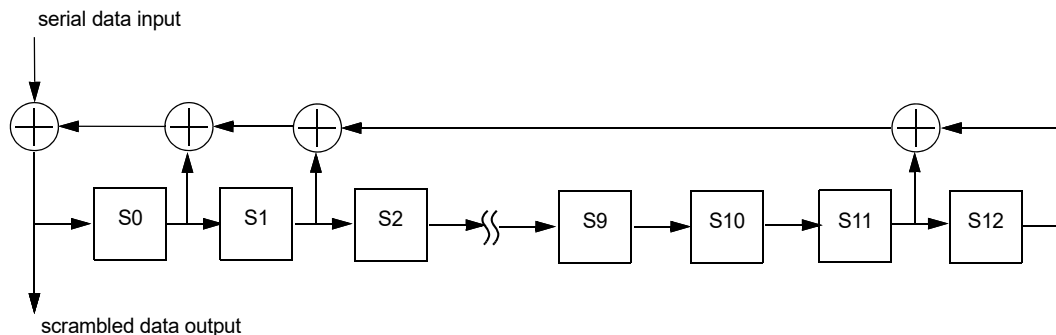


Figure 177-9—Scrambler

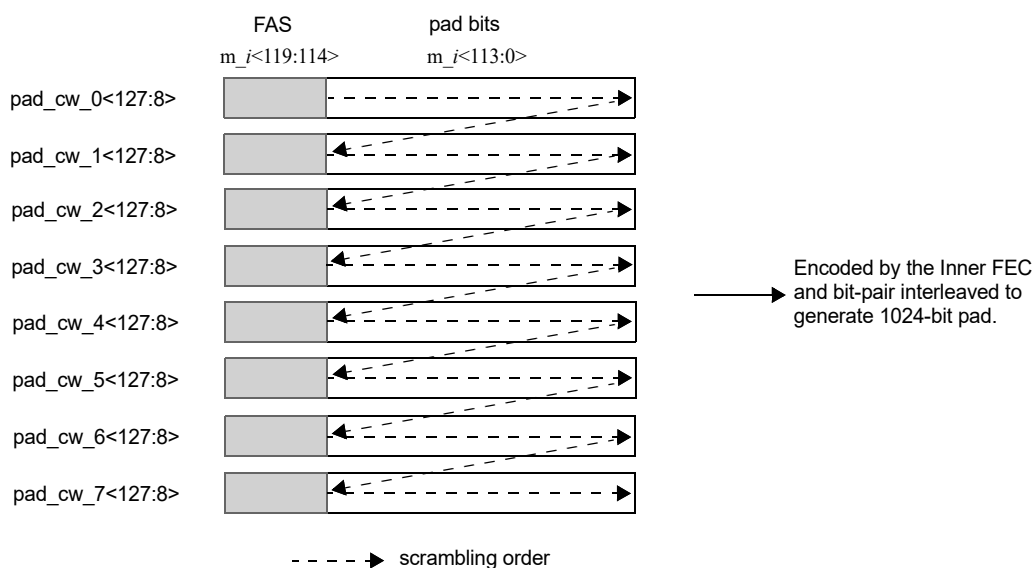


Figure 177-10—Pad bits scrambling order

177.4.7.3 Parity bits

The parity bits of a pad codeword are generated in the same way as for a payload codeword, as defined in 177.4.5.

177.4.7.4 Bit-pair interleaving

The 8 pad codewords are bit-pair interleaved as described in 177.4.6.

177.4.8 PAM4 encoding

The PAM4 encoding function includes Gray mapping as specified in 177.4.8.1 and $1/(1+D) \bmod 4$ precoding as specified in 177.4.8.2.

177.4.8.1 Gray mapping

The Gray mapping for PAM4 encoded lanes is implemented as specified for output lanes in 120.5.7.1.

177.4.8.2 Precoding

The Inner FEC shall provide $1/(1+D) \bmod 4$ precoding capability on each transmit lane. Precoding is implemented as specified for output lanes in 135.5.7.2.

Precoding is enabled and disabled using variables `precoder_tx_out_enable_i` (where i is in the range 0 to 7). If training is enabled by the management variable `mr_training_enable` (see 178B.12), then `precoder_tx_out_enable_i` shall be set as determined by the inter-sublayer link training (ILT) function in the `PATH_READY` state on lane i (see Figure 178B–10a and Figure 178B–10b). The method by which the ILT function affects this variable is implementation dependent. If training is disabled by the management variable `mr_training_enable`, `precoder_tx_out_enable_i` is set as required by the implementation.

177.5 Receive function

The following processes are performed independently on each PMD service interface input lane.

177.5.1 PAM4 decoding

The PAM4 decoding function includes inverse $1/(1+D) \bmod 4$ precoding as specified in 177.5.1.1 and inverse Gray mapping as specified in 177.5.1.2. The hard-decision PAM4 decoding function is depicted as a separate branch for synchronization and test pattern checkers in Figure 177–2. The soft-decision PAM4 decoding used for Inner FEC decoding process may be implemented anywhere in the receive function providing the net behavior is the same.

177.5.1.1 Inverse precoding

The Inner FEC may optionally provide inverse $1/(1+D) \bmod 4$ precoding capability on each receive lane.

If inverse precoding is implemented, it is enabled or disabled as determined by the implementation.

If inverse precoding is enabled, the Inner FEC receive function processes the detected data equivalent to the process specified for input lanes in 135.5.7.2.

If training is enabled by the control variable `mr_training_enable` (see 178B.12), precoding of the received data is enabled at the peer interface transmitter as requested by the receiver using ILT. If training is disabled, then the precoding of data at the peer interface transmitter is controlled by a management entity.

NOTE— If inverse precoding operation is enabled as part of the PAM4 decoding, it also affects the decoding operation in 177.5.5, which does not use the output of the PAM4 decoding function.

177.5.1.2 Inverse Gray mapping

The inverse Gray mapping for PAM4 encoded lanes is identical to the process specified for input lanes in 120.5.7.1.

177.5.2 Inner FEC synchronization

On the receive side, the boundary of Inner FEC codewords shall be located before further processing is done. The eight codewords inserted as pad (see 177.4.7) are then identified and removed before the received data is processed further (see 177.5.3).

Inner FEC synchronization may be performed on PAM4 decoded bit-pair stream using hard-decision decoding. 1:8 bit-pair deinterleaving (each pair of bits corresponding to a PAM4 symbol) is performed to eight Inner FEC flows. The initial position is not specified. Note that this may not be the correct position to

perform the deinterleaving, as the first bit-pair after the deinterleaving position may not be on Inner FEC flow 0.

The Inner FEC achieves codeword synchronization as specified by the Inner FEC codeword synchronization state diagram (Figure 177–13). The synchronization process searches for a valid set of codewords in a window of 128-bit blocks, marking the boundary of these codewords as a candidate location. A candidate location is confirmed as the synchronization location if it is valid for a second window of 128-bit blocks. The synchronization process continuously validates the synchronization location during operation.

After all eight Inner FEC flows have successfully synchronized, Inner FEC flow 0 is identified based on the relative position of Inner FEC codeword boundaries in the synchronized Inner FEC flows. If the blind 1:8 bit-pair deinterleaving (each pair of bits corresponding to a PAM4 symbol) was performed on the correct location, the eight obtained Inner FEC codeword boundaries will be fully aligned, and the flow containing the first bit-pair (corresponding to one PAM4 symbol) after 1:8 bit-pair deinterleaving will be Inner FEC flow 0, as shown in Figure 177–11. Otherwise, there will be one bit-pair of skew between the eight Inner FEC flows, and Inner FEC flow 0 will be the one with a leading bit-pair (corresponding to one PAM4 symbol).

The Inner FEC codeword boundaries found by synchronization are used for deinterleaving of soft inputs into the Inner FEC decoding as described in 177.5.5.

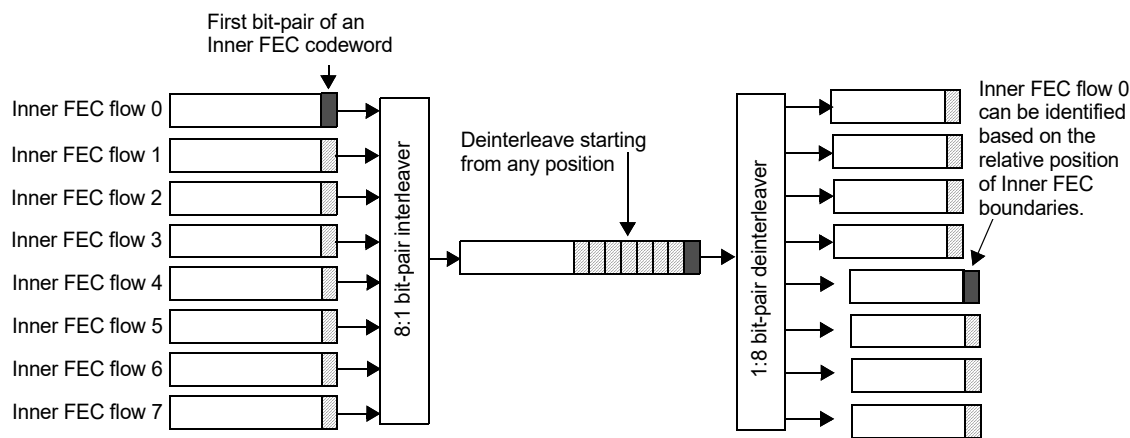


Figure 177–11—Identify Inner FEC flow 0

177.5.3 Pad removal

Once Inner FEC flow 0 is identified, Inner FEC synchronization is achieved. Pad may then be searched based on the codeword boundaries and removed. The search of pad follows the same rules as AM common marker search defined in 119.2.6.2.2, that the 48-bit FAS is compared on a nibble-wise basis (12 comparisons). If nine or more nibbles in the candidate block match the FAS, the candidate block is considered a valid FAS. If two additional valid FASs (total of three consecutive FASs) are successfully located in the next two candidate positions each 8704 codewords away, FAS lock is achieved. The FAS may be optionally monitored, in which case if twelve consecutive FAS do not match the expected values, the FAS detect process may restart.

The state diagram of the FAS lock process is shown in Figure 177–14.

177.5.4 1:8 PAM4 symbol deinterleaving

The received data stream shall be deinterleaved in PAM4 symbols and distributed back to eight Inner FEC flows in a round-robin fashion. The first PAM4 symbol after the pad shall be sent to Inner FEC flow 0.

177.5.5 Inner FEC decode

The Inner FEC decoder is a soft-decision decoder that requires a higher resolution than four values for each received PAM4 symbol. The resolution is implementation specific and is beyond the scope of this standard. The decoder evaluates the incoming block of 64 rx_symbol inputs and determines the most likely payload value. Parity bits are then removed from each Inner FEC codeword.

The decoder should correct all codewords with one bit error. It may also be able to correct codewords with more than one bit error, with a chance of miscorrection.

The following counters shall be implemented per Inner FEC lane:

Inner_FEC_corrected_cw_counter

A 32-bit counter that counts once for each corrected Inner FEC codeword when Inner_FEC_sync_status is true. A corrected Inner FEC codeword is a codeword that contained errors and was corrected or miscorrected. Note that when there is more than one bit error in a codeword there is a chance that the soft decision decoder could miscorrect the codeword. The Inner FEC decoder interprets miscorrected codewords as corrected codewords.

Inner_FEC_uncorrected_cw_counter

A 32-bit counter that counts once for each uncorrected Inner FEC codeword processed when Inner_FEC_sync_status is true. An uncorrected Inner FEC codeword is a codeword that contains errors that the decoder was unable to correct.

Inner_FEC_total_bits_counter

A 64-bit counter that counts once for each bit processed by the Inner FEC decoder when Inner_FEC_sync_status is true. This may be used together with the Inner_FEC_corrected_bits_counter to roughly measure the pre-Inner-FEC BER.

Inner_FEC_corrected_bits_counter

A 64-bit counter that counts once for each bit modified by the Inner FEC decoder when Inner_FEC_sync_status is true. This may be used together with the Inner_FEC_total_bits_counter to roughly measure the pre-Inner-FEC BER.

Inner_FEC_cw_counter

A 48-bit counter that counts once for each Inner FEC codeword received when Inner_FEC_sync_status is true.

Inner_FEC_codeword_error_bin_k

A set of four 32-bit counters, where $k = 0$ to 3. When Inner_FEC_sync_status is true, for $k = 0$ to 2, Inner_FEC_codeword_error_bin_k counts once for each codeword received with exactly k bits detected in error. Inner_FEC_codeword_error_bin_3 increments when Inner_FEC_sync_status is true and three or more bits in an Inner FEC codeword are detected in error and are corrected. For example, if an Inner FEC codeword has exactly two bits detected in error, then Inner_FEC_codeword_error_bin_2 is incremented. Note that Inner_FEC_codeword_error_bin_0 is only incremented for codewords determined to be correct by the decoder (zero detected incorrect bits) and is not incremented for codewords determined to be uncorrectable which are counted in Inner_FEC_uncorrected_cw_counter.

For each codeword processed, exactly one counter in Inner_FEC_codeword_error_bin_k or Inner_FEC_uncorrected_cw_counter is incremented.

177.5.6 Inverse circular shift

The inverse circular shift shall perform the inverse of the circular shift function (see 177.4.4) to restore the order of 12 RS-FEC symbols in each decoded codeword.

177.5.7 8:1 120-bit block multiplexer

After the inverse circular shift the eight Inner FEC flows shall be multiplexed into one stream on a 120-bit block (three RS-FEC symbol-quartet) basis in a round-robin fashion from Inner FEC flow 0 through 7 before being fed into the convolutional deinterleaver. This provides the inverse function of 177.4.3.

177.5.8 Convolutional deinterleaver

The convolutional deinterleaver shall perform the inverse function of the convolutional interleaver function (see 177.4.2) to restore the order of the data received to be compatible with the BASE-R SM-PMA.

The convolutional deinterleaver contains three delay lines. The first line (Delay Line 0) adds no delay, the second line (Delay Line 1) delays the data by $1 \times Q \times 4$ RS-FEC symbols, and the last line (Delay Line 2) by $2 \times Q \times 4$ RS-FEC symbols. The values of Q are shown in Table 177-3. RS-FEC Symbol-quartets from the 120-bit block multiplexer are distributed sequentially to delay line 0, delay line 1, and delay line 2. The first 40-bit RS-FEC symbol-quartet from each 120-bit block is always aligned to delay line 0.

The output of the convolutional deinterleaver round-robins between the three delay lines, sending one RS-FEC symbol-quartet from each at a time, in the order Delay Line 0, then Delay Line 1, and lastly Delay Line 2.

Figure 177-12 illustrates the delay lines of the convolutional deinterleaver using Q instances of a RS-FEC symbol-quartet sized base delay element labeled “D”. The input and output switches are always aligned to the same row.

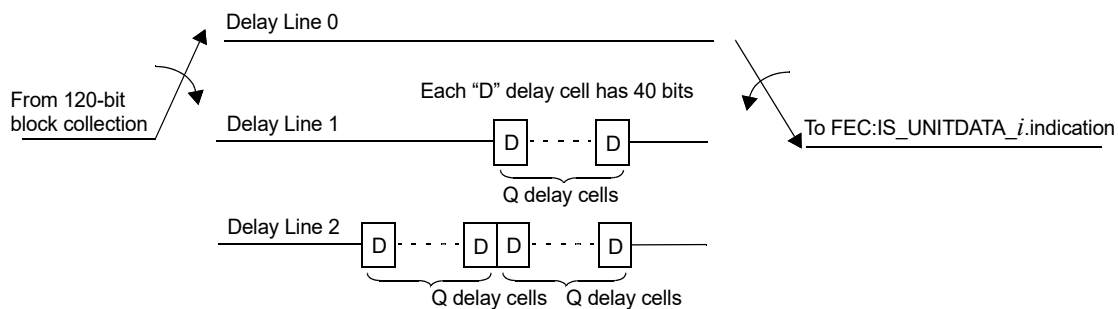


Figure 177-12—Convolutional deinterleaver

177.6 Test patterns

177.6.1 Test pattern generation

The Inner FEC includes test pattern generators. All test pattern generators defined in this subclause are per Inner FEC lane. These test patterns are used to drive each transmit output lane towards the PMD service interface for PMD testing purposes.

Enabling any of the pattern generators on an Inner FEC lane only affects that lane. The behavior when more than one generator is enabled on the same lane is not specified.

177.6.1.1 PRBS31 encoded by Inner FEC test pattern

The Inner FEC shall support a test mode, where a PRBS31 test pattern generated from each transmit output lane i of the PMA above the Inner FEC (see 176.7.4.1) is processed by the Inner FEC sublayer, with the exception that the alignment lock and deskew function on the Inner FEC transmit path (see 177.4.1) is disabled and bypassed.

This test mode is enabled on each Inner FEC lane i using the PRBS31_encoded_by_Inner_FEC_ i management variable and enabling the PRBS31 generator on each transmit output lane i of the PMA above the Inner FEC.

177.6.1.2 PRBS31 test pattern

The Inner FEC shall include a PRBS31 test-pattern generator, as specified in 120.5.11.1.1, at the input to the PAM4 encoder.

177.6.1.3 PRBS13Q test pattern

The Inner FEC may include a PRBS13Q test-pattern generator, as specified in 120.5.11.2.1.

177.6.1.4 PRBS31Q test pattern

The PRBS31Q test pattern generation is provided by the PRBS31 generator (see 177.6.1.2), followed by Gray mapping, and precoding (if enabled) in the PAM4 encoder (see 177.4.8).

177.6.1.5 SSPRQ test pattern

The Inner FEC shall include a short stress pattern random quaternary (SSPRQ) test-pattern generator, as specified in 120.5.11.2.3.

177.6.1.6 Square wave (quaternary) test pattern

The Inner FEC may support a square wave (quaternary) test-pattern generator, as specified in 120.5.11.2.4.

177.6.2 Test pattern checking

The Inner FEC includes test pattern checkers. All test pattern checkers defined in this subclause are per Inner FEC lane. These test pattern checkers are used to check the signal on each receive input lane from the PMD service interface for PMD testing purposes.

177.6.2.1 PRBS31 test pattern checker

The Inner FEC shall include test pattern checkers for PRBS31, as specified in 120.5.11.1.1, at the output of the PAM4 decoder.

177.6.2.2 PRBS31Q test pattern checking

For PRBS31Q checking, the hard-decision PAM4 decoder (see 177.5.1) provides inverse precoding (if enabled) and inverse Gray mapping and the PRBS31 checker (see 177.6.2.1) checks for bit errors.

177.6.2.3 PRBS31 encoded by Inner FEC test pattern checker

The Inner FEC shall support a test mode, where a PRBS31 test pattern is encoded by the Inner FEC (see 176.7.4). For this test mode, the PRBS31 pattern is recovered by the Inner FEC receive function on each Inner FEC lane i and the PRBS31 checkers in the PMA above the Inner FEC on each receive input lane i (see 176.7.4.1) are used to check the pattern.

177.7 State diagrams

This subclause is composed of state diagrams, including the associated definitions of constants, variables, functions, and counters.

177.7.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

177.7.2 State diagram definitions

177.7.2.1 Variables

all_synced

A Boolean variable that is set to true when `sync_flow<x>` is true for all eight flows and the Inner FEC flow 0 is identified, and is false otherwise.

fas_cnt_done

Boolean variable that is set to true when the counter `fas_cnt` has reached its terminal count and is set to false when starting the counter (see Figure 177–14).

fas_lock

A Boolean variable that indicates the receiver has detected the location of the frame alignment sequence within the pad codewords. Its value is set by the Inner FEC pad detection state diagram (Figure 177–14).

fas_valid

A Boolean variable that is set to true if the received 48-bit block is a valid frame alignment sequence payload. Otherwise, this variable is set to false. The frame alignment sequence payload is described in 177.4.7. The 48 bits are compared on a nibble-wise basis (12 comparisons). If nine or more nibbles in the candidate block match the corresponding nibbles in the sequence defined in 177.4.7, the candidate block is considered a valid frame alignment sequence payload.

FEC_reset

A Boolean variable that is true when set by a management entity and is false otherwise.

Inner_FEC_sync_status

A Boolean variable that is set to true when `all_synced` and `fas_lock` are both true and is set to false otherwise.

reset

A Boolean variable that controls the resetting of the Inner FEC sublayer. It is true whenever a reset is necessary including when initiated by `FEC_reset` and during power on, and is false otherwise.

restart_inner_fec_sync

A Boolean variable that is used to restart all eight self-synchronization processes as well as the pad detection process associated with an input lane in the receive direction. Its value can be set to true in either the Inner FEC self-synchronization state diagram (Figure 177–13) or the Inner FEC pad

detection state diagram (Figure 177–14). Its value is set to false upon entering the FAS_LOCK_INIT state of the Inner FEC pad detection state diagram.

signal_ok

A Boolean variable that is set based on the most recently received SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive. It is true if the value is OK and false otherwise.

slip_done

A Boolean variable that indicates the next candidate 128-bit block position can be tested by the Inner FEC self-synchronization process. It is set to true when the SLIP function completes and is set to false upon entering the GET_BLOCK state of the Inner FEC self-synchronization state diagram (Figure 177–13).

sync_flow<*x*>

A Boolean variable that indicates the Inner FEC codeword boundary is found for an Inner FEC flow, where *x* = 0 to 7, and *x* represents an Inner FEC flow ID before identifying the actual Inner FEC flow numbering. The value of sync_flow<*x*> is set by the Inner FEC self-synchronization state diagram (Figure 177–13).

test_cw

A Boolean variable that is set to true when a new Inner FEC codeword is available for decoding and is set to false when the CW_CHECK_*n* state is entered, where *n* = 1 to 3.

test_fas

A Boolean variable that is set to true when a candidate block position is available for testing and false when the FIND_1ST state is entered.

valid_cw

A Boolean variable that is set to true when the calculated syndrome of the Inner FEC codeword being tested is zero and is set to false otherwise.

177.7.2.2 Functions

SLIP

Causes the next candidate block position to be tested. The precise method for determining the next candidate block position is not specified and is implementation dependent. However, an implementation shall evaluate all possible block positions.

177.7.2.3 Counters

bad_cw_cnt

Counts the number of invalid Inner FEC codewords received within a period of 150 codewords. An invalid Inner FEC codeword is indicated when valid_cw is false.

bad_fas_cnt

Counts the number of consecutive FAS that do not match the expected values.

cw_cnt

Counts the number of Inner FEC codewords processed.

fas_cnt

Counts the interval of Inner FEC codewords between two adjacent pads as defined in 177.4.7.

valid_cw_cnt

Counts the number of valid Inner FEC codewords within a period of 50 codewords. A valid Inner FEC codeword is indicated when valid_cw is true.

177.7.3 State diagram figures

The Inner FEC sublayer shall implement eight self-synchronization processes, as shown in Figure 177–13, for each input lane in the receive direction. Each synchronization process operates independently on an Inner FEC flow to identify the boundaries of the Inner FEC codewords.

An Inner FEC Pad detection process as illustrated in the state diagram in Figure 177–14 shall be implemented for each input lane in the receive direction.

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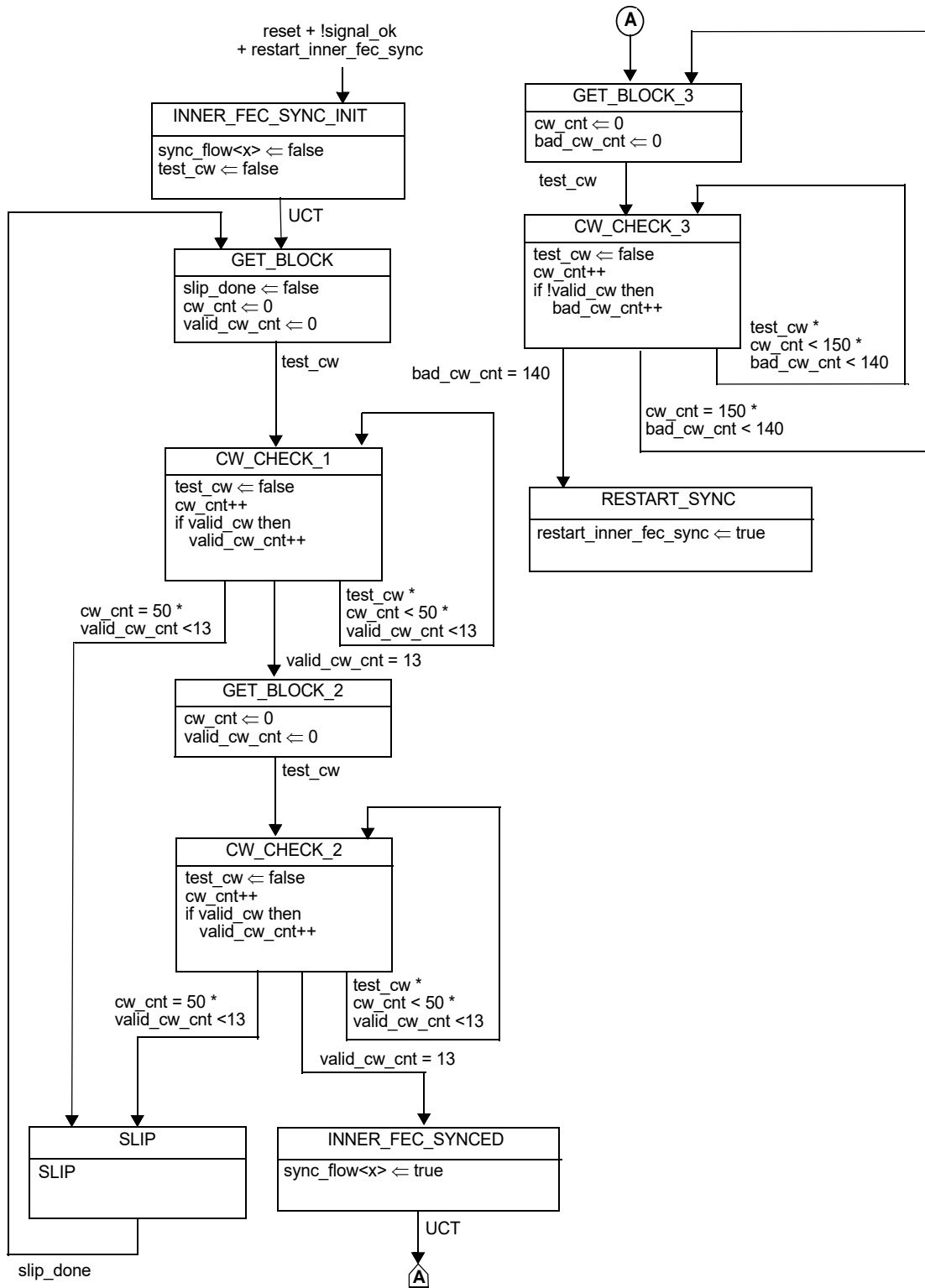


Figure 177–13—Inner FEC self-synchronization state diagram

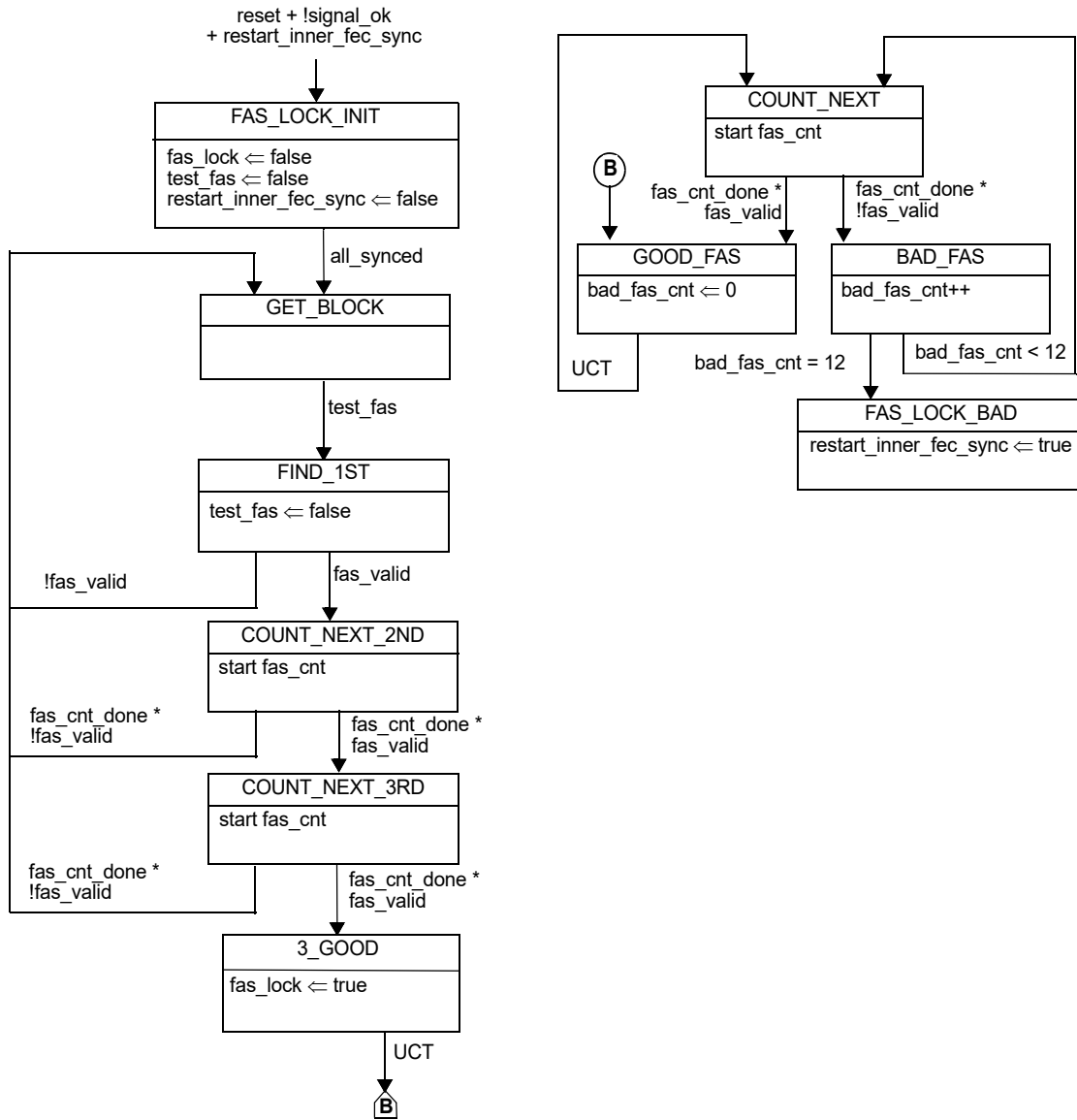


Figure 177-14—Inner FEC pad detection state diagram

177.8 Delay constraints

The maximum delay contributed by the Inner FEC (sum of transmit and receive delays at one end of the link) shall be no more than the values provided in Table 177–6. A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4, 169.4, 174.4, and their references.

Table 177–6—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
200GBASE-R Inner FEC	59 392	116	296.96
400GBASE-R Inner FEC	70 656	138	176.64
800GBASE-R Inner FEC	93 184	182	116.48
1.6TBASE-R Inner FEC	138 240	270	86.4

177.9 Path data delay for time synchronization

The Inner FEC sublayer optionally provides path data delay information for time synchronization (see Clause 90).

The ability to report Inner FEC path data delays is indicated by the Inner_FEC_delay_ns_TX_ability, Inner_FEC_delay_subns_TX_ability, Inner_FEC_delay_ns_RX_ability, and Inner_FEC_delay_subns_RX_ability status variables (see Table 177–8).

The Inner FEC sublayer has an intrinsic delay pattern where the sum of the transmit and receive delays is a constant value. As recommended in 90A.7, the maximum delay value is allocated to the transmit path and the minimum delay value to the receive path. The transmit and receive path data delays are reported as the delay incurred by the first symbol on FEC flow 0 after the 1024-bit pad insertion (see 177.4.7).

Four separate delays are reported in the following eight path data delay status variables:

- Inner_FEC_delay_ns_TX_max, Inner_FEC_delay_subns_TX_max
- Inner_FEC_delay_ns_TX_min, Inner_FEC_delay_subns_TX_min
- Inner_FEC_delay_ns_RX_max, Inner_FEC_delay_subns_RX_max
- Inner_FEC_delay_ns_RX_min, Inner_FEC_delay_subns_RX_min

177.10 Management variables

Inner FEC control and status variables intended to be accessible via a management system are listed in Table 177–7 and Table 177–8.

If the MDIO interface (see Clause 45) is implemented, Inner FEC control and status variables shall be mapped to MDIO registers as shown in Table 177–7 and Table 177–8.

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 177–7—Inner FEC control variables and MDIO mapping

Control variable	Variable reference	MDIO Register/bit number	MDIO register/bit reference
FEC_reset	177.7.2.1	1.0.15	45.2.1.1
precoder_tx_out_enable<0:7>	177.4.8.2	1.600.0:7	45.2.1.139
PRBS31_encoded_by_Inner_FEC<0:7>	177.6.1.1	1.2405.0:7	45.2.1.255
PRBS13Q_pattern_enable<0:7>	177.6.1.3	1.2401.0:7	45.2.1.251
PRBS31_pattern_enable<0:7>	177.6.1.2	1.2402.0:7	45.2.1.252
SSPRQ_pattern_enable<0:7>	177.6.1.5	1.2403.0:7	45.2.1.253
Square_wave_enable<0:7>	177.6.1.6	1.2404.0:7	45.2.1.254

Table 177–8—Inner FEC status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
pmal_locked_demux, Inner FEC lane 0 through 7	176.4.4.2.1	1.2410.8:15	45.2.1.256
Inner_FEC_sync_status, Inner FEC lane 0 through 7	177.7.2.1	1.2410.0:7	45.2.1.256
Inner_FEC_corrected_cw_counter (Inner FEC lane 0)	177.5.5	1.2412, 1.2413	45.2.1.258
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 0)	177.5.5	1.2414, 1.2415	45.2.1.259
Inner_FEC_total_bits_counter (Inner FEC lane 0)	177.5.5	1.2416, 1.2417, 1.2418, 1.2419	45.2.1.260
Inner_FEC_corrected_bits_counter (Inner FEC lane 0)	177.5.5	1.2420, 1.2421, 1.2422, 1.2423	45.2.1.261
Inner_FEC_cw_counter (Inner FEC lane 0)	177.5.5	1.2424, 1.2425, 1.2426	45.2.1.262
Inner_FEC_codeword_error_bin_k (Inner FEC lane 0)	177.5.5	1.2427 through 1.2434	45.2.1.263
Inner_FEC_corrected_cw_counter (Inner FEC lane 1)	177.5.5	1.2437, 1.2438	45.2.1.264
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 1)	177.5.5	1.2439, 1.2440	45.2.1.264
Inner_FEC_total_bits_counter (Inner FEC lane 1)	177.5.5	1.2441, 1.2442, 1.2443, 1.2444	45.2.1.264

Table 177–8—Inner FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Inner_FEC_corrected_bits_counter (Inner FEC lane 1)	177.5.5	1.2445, 1.2446, 1.2447, 1.2448	45.2.1.264
Inner_FEC_cw_counter (Inner FEC lane 1)	177.5.5	1.2449, 1.2450, 1.2451	45.2.1.264
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 1)	177.5.5	1.2452 through 1.2459	45.2.1.264
Inner_FEC_corrected_cw_counter (Inner FEC lane 2)	177.5.5	1.2462, 1.2463	45.2.1.264
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 2)	177.5.5	1.2464, 1.2465	45.2.1.264
Inner_FEC_total_bits_counter (Inner FEC lane 2)	177.5.5	1.2466, 1.2467, 1.2468, 1.2469	45.2.1.264
Inner_FEC_corrected_bits_counter (Inner FEC lane 2)	177.5.5	1.2470, 1.2471, 1.2472, 1.2473	45.2.1.264
Inner_FEC_cw_counter (Inner FEC lane 2)	177.5.5	1.2474, 1.2475, 1.2476	45.2.1.264
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 2)	177.5.5	1.2477 through 1.2484	45.2.1.264
Inner_FEC_corrected_cw_counter (Inner FEC lane 3)	177.5.5	1.2487, 1.2488	45.2.1.264
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 3)	177.5.5	1.2489, 1.2490	45.2.1.264
Inner_FEC_total_bits_counter (Inner FEC lane 3)	177.5.5	1.2491, 1.2492, 1.2493, 1.2494	45.2.1.264
Inner_FEC_corrected_bits_counter (Inner FEC lane 3)	177.5.5	1.2495, 1.2496, 1.2497, 1.2498	45.2.1.264
Inner_FEC_cw_counter (Inner FEC lane 3)	177.5.5	1.2499, 1.2500, 1.2501	45.2.1.264
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 3)	177.5.5	1.2502 through 1.2509	45.2.1.264
Inner_FEC_corrected_cw_counter (Inner FEC lane 4)	177.5.5	1.2512, 1.2513	45.2.1.264

Table 177–8—Inner FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 4)	177.5.5	1.2514, 1.2515	45.2.1.264
Inner_FEC_total_bits_counter (Inner FEC lane 4)	177.5.5	1.2516, 1.2517, 1.2518, 1.2519	45.2.1.264
Inner_FEC_corrected_bits_counter (Inner FEC lane 4)	177.5.5	1.2520, 1.2521, 1.2522, 1.2523	45.2.1.264
Inner_FEC_cw_counter (Inner FEC lane 4)	177.5.5	1.2524, 1.2525, 1.2526	45.2.1.264
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 4)	177.5.5	1.2527 through 1.2534	45.2.1.264
Inner_FEC_corrected_cw_counter (Inner FEC lane 5)	177.5.5	1.2537, 1.2538	45.2.1.264
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 5)	177.5.5	1.2539, 1.2540	45.2.1.264
Inner_FEC_total_bits_counter (Inner FEC lane 5)	177.5.5	1.2541, 1.2542, 1.2543, 1.2544	45.2.1.264
Inner_FEC_corrected_bits_counter (Inner FEC lane 5)	177.5.5	1.2545, 1.2546, 1.2547, 1.2548	45.2.1.264
Inner_FEC_cw_counter (Inner FEC lane 5)	177.5.5	1.2549, 1.2550, 1.2551	45.2.1.264
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 5)	177.5.5	1.2552 through 1.2559	45.2.1.264
Inner_FEC_corrected_cw_counter (Inner FEC lane 6)	177.5.5	1.2562, 1.2563	45.2.1.264
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 6)	177.5.5	1.2564, 1.2565	45.2.1.264
Inner_FEC_total_bits_counter (Inner FEC lane 6)	177.5.5	1.2566, 1.2567, 1.2568, 1.2569	45.2.1.264
Inner_FEC_corrected_bits_counter (Inner FEC lane 6)	177.5.5	1.2570, 1.2571, 1.2572, 1.2573	45.2.1.264
Inner_FEC_cw_counter (Inner FEC lane 6)	177.5.5	1.2474, 1.2575, 1.2576	45.2.1.264

Table 177–8—Inner FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Inner_FEC_codeword_error_bin_k (Inner FEC lane 6)	177.5.5	1.2577 through 1.2584	45.2.1.264
Inner_FEC_corrected_cw_counter (Inner FEC lane 7)	177.5.5	1.2587, 1.2588	45.2.1.264
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 7)	177.5.5	1.2589, 1.2590	45.2.1.264
Inner_FEC_total_bits_counter (Inner FEC lane 7)	177.5.5	1.2591, 1.2592, 1.2593, 1.2594	45.2.1.264
Inner_FEC_corrected_bits_counter (Inner FEC lane 7)	177.5.5	1.2595, 1.2596, 1.2597, 1.2598	45.2.1.264
Inner_FEC_cw_counter (Inner FEC lane 7)	177.5.5	1.2599, 1.2600, 1.2601	45.2.1.264
Inner_FEC_codeword_error_bin_k (Inner FEC lane 7)	177.5.5	1.2602 through 1.2609	45.2.1.264
PRBS13Q_gen_Tx_ability	177.6.1.3	1.1500.11	45.2.1.169
Square_wave_ability	177.6.1.6	1.1500.12	45.2.1.169
FEC_delay_ns_TX_max	177.9	1.1813, 1.1814	45.2.1.177a
FEC_delay_subns_TX_max	177.9	1.1817	45.2.1.177a
FEC_delay_ns_TX_min	177.9	1.1815, 1.1816	45.2.1.177a
FEC_delay_subns_TX_min	177.9	1.1818	45.2.1.177a
FEC_delay_ns_RX_max	177.9	1.1819, 1.1820	45.2.1.177b
FEC_delay_subns_RX_max	177.9	1.1823	45.2.1.177b
FEC_delay_ns_RX_min	177.9	1.1821, 1.1822	45.2.1.177b
FEC_delay_subns_RX_min	177.9	1.1824	45.2.1.177b
FEC_delay_ns_RX_ability	177.9	1.1800.4	45.2.1.175
FEC_delay_subns_RX_ability	177.9	1.1800.5	45.2.1.175
FEC_delay_ns_TX_ability	177.9	1.1800.6	45.2.1.175
FEC_delay_subns_TX_ability	177.9	1.1800.7	45.2.1.175

177.11 Protocol implementation conformance statement (PICS) proforma for Clause 177, Inner FEC sublayer, type 200GBASE-R, 400GBASE-R, 800GBASE-R and 1.6TBASE-R

177.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 177, Inner FEC, type 200GBASE-R, 400GBASE-R, 800GBASE-R and 1.6TBASE-R shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

177.11.2 Identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

177.11.2.1 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 177, Inner FEC, type 200GBASE-R, 400GBASE-R, 800GBASE-R and 1.6TBASE-R
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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177.11.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MD	MDIO	177.10		O	Yes [] No []

177.11.4 PICS proforma tables for Inner FEC sublayer, type 200GBASE-R, 400GBASE-R, 800GBASE-R and 1.6TBASE-R

177.11.4.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Symbol demultiplexing and deskew	177.4.1		M	Yes []
TF2	Convolutional interleaver	177.4.2		M	Yes []
TF3	120-bit block distribution	177.4.3		M	Yes []
TF4	Circular shift	177.4.4		M	Yes []
TF5	Inner FEC encode	177.4.5		M	Yes []
TF6	8:1 bit-pair interleaving	177.4.6		M	Yes []
TF7	Pad insertion and format	177.4.7		M	Yes []
TF8	PRBS31-encoded-by-Inner-FEC TX mode	177.6.1.1		M	Yes []
TF9	PRBS31 test pattern generator	177.6.1.2		M	Yes []
TF10	SSPRQ generator	177.6.1.5		M	Yes []
TF11	PRBS31 checker	177.6.2.1		M	Yes []

177.11.4.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
RF1	Inner FEC synchronization and pad removal	177.5.2		M	Yes []
RF2	1:8 PAM4 deinterleaving	177.5.4		M	Yes []
RF3	Inner FEC decode	177.5.5		M	Yes []
RF4	Inverse circular shift	177.5.6		M	Yes []
RF5	120-bit block multiplex	177.5.7		M	Yes []
RF6	Convolutional deinterleaver	177.5.8		M	Yes []
RF7	PRBS31-encoded-by-Inner-FEC RX mode	177.6.2.3		M	

177.11.4.3 State diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
SM1	Inner FEC synchronization process	177.7.3		M	Yes []
SM2	Pad lock process	177.7.3		M	Yes []

177.11.4.4 Delay constraints

Item	Feature	Subclause	Value/Comment	Status	Support
TIM1	Delay constraints	177.8		M	Yes []

177.11.4.5 Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	177.9		O	Yes [] No []
TS1	Path data delays	177.9	Path data delays reported as if the DDMP is at the first symbol on FEC flow 0 after the 1024-bit pad insertion	TS:M	Yes [] N/A []
TS2	Nanosecond resolution	177.9	Path data delays reported with nanosecond or better resolution.	TS:M	Yes [] N/A []
TS3	Sub-nanosecond resolution	177.9	Path data delays reported with sub-nanosecond resolution.	TS:O	Yes [] No [] N/A []

177.11.4.6 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	177.10		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	177.10		MD:M	Yes [] N/A []